Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started
- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra
- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations
- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices
- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces
- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • Grade: 85.71%

Lesson 5: Orthogonal Complement and Projection
- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min
- Importance of Projections
  Reading • 4 min
- Projection Onto One-Dimensional Subspaces (Lines)
  Reading • 15 min
- Projection Onto General Subspaces
  Reading • 10 min
- **Projections Case Study**
  Reading • 10 min

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Projections Case Study

0:24 / 3:40 2x

Imagine we're analyzing data from a fitness study that records two features:

1. Workout time (hours per week)
2. Calories burned (in thousands per week)

We have a dataset that lives in 3D space, including an additional feature: 3. Weight loss (in kg per week).

We are given:

$$\mathbf{x} = \begin{bmatrix} 6 \\ 3 \\ 4 \end{bmatrix}$$

represents an individual's workout time, calories burned and weight loss,

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

a basis vector capturing a relationship between workout time and calories burned (For every 1 unit increase in workout time, the trend shows a 2-unit increase in calories burned), and

$$\mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

a basis vector capturing a relationship between calories burned and weight loss (For every 1-unit increase in calories burned, there is a 1-unit increase in weight loss, according to this vector).

The two basis vectors $\mathbf{b}_1 and \mathbf{b}_2$ span a $2D$ subspace. Our goal is to:

1. Project $\mathbf{x}$ onto this subspace.
2. Compute the error vector, representing the portion of $\mathbf{x}$ not explained by the subspace.

Solution:

1. Matrix of Basis Vectors:

The subspace is spanned by $\mathbf{b}_1 and \mathbf{b}_2$. Combine them into a matrix B:

$$B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

2. Projection Matrix:

The projection matrix P is:

$$P = B(B^T B)^{-1} B^T$$

a. Compute $B^T B$:

$$B^T B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}.$$

b. Compute $(B^T B)^{-1}$

The determinant of $B^T B$:

$$det(B^T B) = (5)(2) - (2)(2) = 6.$$

The Inverse is:

$$(B^T B)^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}$$

c. Compute P

$$P = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{5}{6} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{6} & \frac{5}{6} \end{bmatrix}$$

3. Projection of $\mathbf{x}$:

$$\pi(\mathbf{x}) = P\mathbf{x} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{5}{6} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{6} & \frac{5}{6} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \\ 4 \end{bmatrix}$$

$$\pi(\mathbf{x}) = \begin{bmatrix} 2.77 \\ 4.08 \\ 4.15 \end{bmatrix}$$

4. Error Vector:

$$\mathbf{x} - \pi(\mathbf{x}) = \begin{bmatrix} 6 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 2.77 \\ 4.08 \\ 4.15 \end{bmatrix} = \begin{bmatrix} 3.23 \\ -1.08 \\ -0.15 \end{bmatrix}$$

Interpretation

Projection of $\mathbf{x}$:

$$\pi(\mathbf{x}) = \begin{bmatrix} 2.77 \\ 4.08 \\ 4.15 \end{bmatrix}$$

This is the portion of $\mathbf{x}$ (workout, calories, weight loss) that align with the overall trend captured by $\mathbf{b}_1 and \mathbf{b}_2$

Error vector:

$$\mathbf{x} - \pi(\mathbf{x}) = \begin{bmatrix} 3.23 \\ -1.08 \\ -0.15 \end{bmatrix}$$

This is the portion of $\mathbf{x}$ that doesn't align with the subspace, representing deviation from the trend.

Go to next item ✓ Completed

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